CMSC730 - Spring 2026
University of Maryland

[Assignment 5]

Radar

Abstract

In this assignment, you will construct a simple radar system using a servo motor, an ultrasonic sensor, and the 3D printed gear you created in the previous project. The goal of this exercise is to combine mechanical movement with sensor-based measurement to simulate the basic principles of a scanning radar. By integrating these components, you will learn how to coordinate motion control with distance sensing and how to visualize data collected from a rotating sensor platform. This project reinforces your understanding of servo control, timing, and sensor data processing in embedded systems.

Description

You will build a functional radar mechanism that uses a servo motor to sweep the ultrasonic sensor across a defined range of angles. As the servo rotates, the ultrasonic sensor will continuously measure the distance to nearby objects within its field of view. The measured distances can then be used to infer the position of obstacles relative to the sensor, creating the effect of a radar that scans its surroundings.

Your mechanical setup should make use of the 3D printed gear from the previous assignment to connect the servo motor and the ultrasonic sensor securely. Ensure that the rotation is smooth and stable so that the sensor readings remain consistent. You are encouraged to test different angles of rotation and scanning speeds to achieve reliable data collection.

As the servo sweeps from one side to the other, record the distance readings obtained from the ultrasonic sensor at regular intervals. You may choose to display these results on a serial monitor or store them for later visualization. The output should clearly indicate how distance values change as the servo angle changes, allowing you to identify where objects are located within the scanned area.

Delivery

There are two deliverables and one optional submission.

- First, your code. You can put them in a zip file if needed.
- Second, a quick video demo of the working radar with an (unlisted) youtube video link. In the video, please put some obstacles in front of your radar and show the distance reading changes from the serial port.

Due Date

Th Mar 26th, 11:59 PM EST

Last Modified Date: Mar 12, 2026